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JDK vs JRE vs JVM — differences & setup explain with example

JVM vs JRE vs JDK

These three are nested components — each builds on the previous one.

JVM — Java Virtual Machine

The **engine** that actually *runs* Java bytecode. It's an abstract machine that:

- Reads `.class` files (compiled bytecode)
- Translates them to native machine code at runtime (via JIT compiler)
- Manages memory (heap, stack, garbage collection)

Key point: JVM is platform-specific (Windows JVM ≠ Linux JVM), but the bytecode it runs is platform-independent — this is the "write once, run anywhere" promise.

JRE — Java Runtime Environment

JVM + standard libraries. Everything needed to *run* a Java program, but not compile one.

Contains:

- JVM
- Core class libraries (`java.lang`, `java.util`, `java.io`, etc.)
- Runtime support files

Use it if you just want to **run** `.jar` or `.class` files.

JDK — Java Development Kit

JRE + development tools. Everything needed to *write and compile* Java programs.

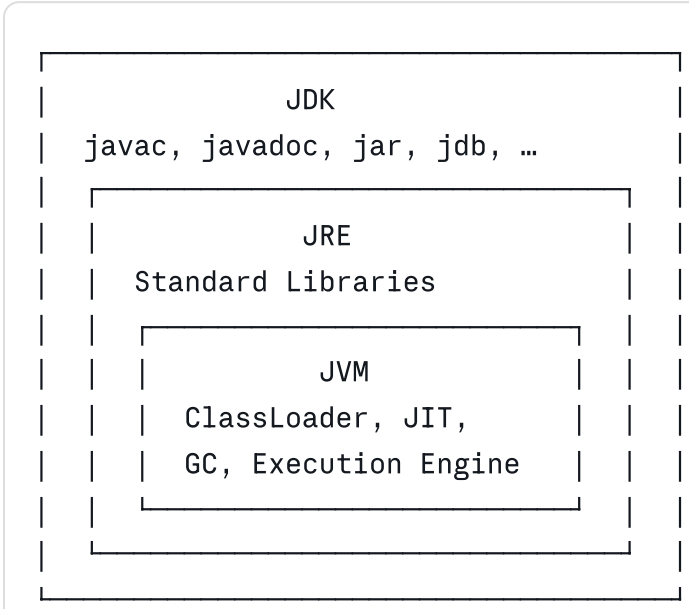
Contains:

- JRE (which includes JVM)

- `javac` — the compiler
- `javadoc`, `jdb` (debugger), `jar`, `jshell`, and more
- Header files for native code

Use it if you want to **build** Java applications.

Visual relationship



End-to-end example

1. You write source code → `Hello.java`

java

```java
public class Hello {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

2. JDK's `javac` compiles it → produces `Hello.class` (bytecode)

bash

```bash
javac Hello.java      # needs JDK
```

3. JVM (via `java`) runs the bytecode

bash

```bash
java Hello            # needs JRE/JVM
# Output: Hello, World!
```

The `.class` file contains bytecode — not machine code. The JVM reads it and compiles it to native instructions at runtime (that's the JIT compiler doing its job).

Quick decision guide

Goal	You need
Run a `.jar` / `.class` file	JRE (or JDK)
Write & compile Java code	JDK
Deploy a Java app on a server	JRE (historically) or JDK
Use an IDE (IntelliJ, Eclipse)	JDK

> **Note:** Since Java 11, Oracle no longer ships a standalone JRE — the JDK is the standard install for everything. Tools like `jlink` let you bundle a minimal custom runtime for deployment.

Setup (quick)

```bash
# macOS (Homebrew)
brew install openjdk@21

# Ubuntu/Debian
sudo apt install openjdk-21-jdk

# Verify
java -version    # shows JVM version
javac -version   # confirms JDK tools are present
```

The most widely used distribution is **OpenJDK** (free, open-source). Others include Amazon Corretto, Eclipse Temurin, and GraalVM.

How Java code compiles and runs (bytecode, classloader)

How Java Code Compiles and Runs

The journey from `.java` source to running program has two distinct phases: **compile time** (done once) and **runtime** (done every execution).

Phase 1 — Compile Time (javac)

```
Hello.java  ──javac──▶  Hello.class
(source)                 (bytecode)
```

`javac` does several things:

- **Lexing & Parsing** — tokenizes source, builds an AST (Abstract Syntax Tree)
- **Semantic analysis** — type checking, resolving symbols
- **Bytecode generation** — emits platform-neutral `.class` file

The `.class` file is **not machine code** — it's an intermediate instruction set designed for the JVM, not your CPU.

bash

```
javac Hello.java       # produces Hello.class
javap -c Hello.class   # disassemble — peek at bytecode
```

Sample `javap` output:

```
public static void main(java.lang.String[]);
  Code:
     0: getstatic     #7   // Field System.out
     3: ldc           #13  // String "Hello, World!"
     5: invokevirtual #15  // Method println
     8: return
```

These are **JVM opcodes** — instructions like `getstatic`, `ldc`, `invokevirtual`. The JVM understands these regardless of OS.

Phase 2 — Runtime (JVM)

Once you run `java Hello`, the JVM takes over in three big steps:

Step 1 — ClassLoader Subsystem

The ClassLoader **finds, loads, links, and initializes** `.class` files on demand (lazy — only when first needed).

Three built-in ClassLoaders (Java 9+):

ClassLoader	Loads
Bootstrap	Core JDK classes (`java.lang.*`, `java.util.*`)
Platform	Extension/platform modules (`javax.*`, etc.)
Application	Your classes + classpath jars

They form a **parent-delegation chain**: before loading a class, each loader asks its parent first. This prevents you from accidentally overriding `java.lang.String`.

```
Your class request
       |
       ▼
Application ClassLoader
  → asks Platform ClassLoader
      → asks Bootstrap ClassLoader
          → found? return it
          → not found? try Platform
              → not found? try Application
                  → not found? ClassNotFoundException
```

Three sub-phases inside loading:

```
Load → Link → Initialize
```

- **Load** — read the `.class` bytes into memory
- **Link**
  - *Verify* — validate bytecode is well-formed & safe
  - *Prepare* — allocate memory for static fields (set to defaults)
  - *Resolve* — replace symbolic references with direct memory references
- **Initialize** — run static initializers, assign static field values

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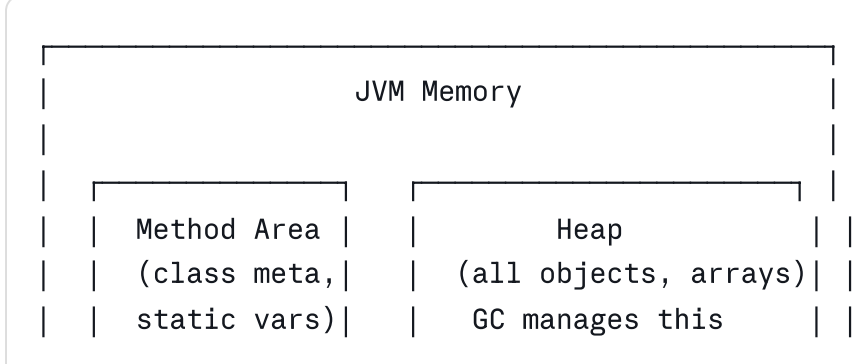
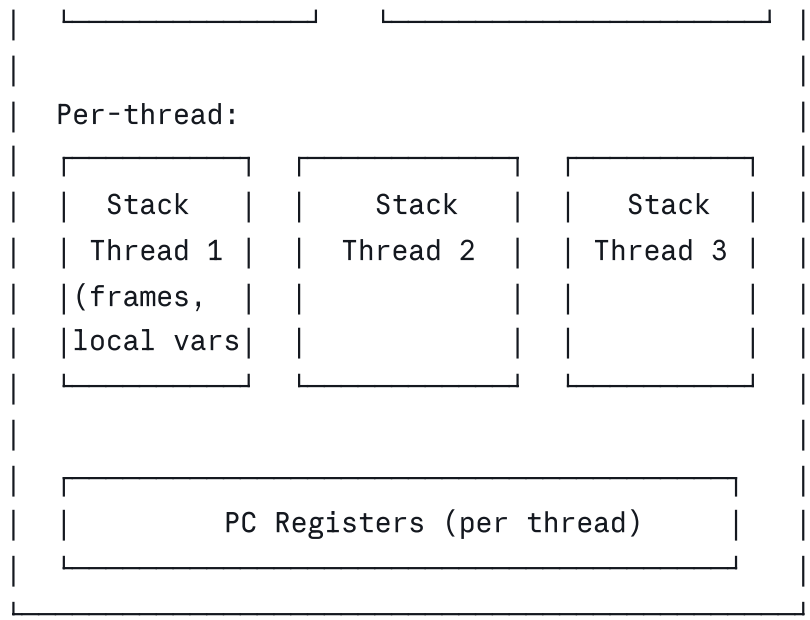
Step 2 — Runtime Data Areas (Memory)

The JVM carves memory into distinct regions:



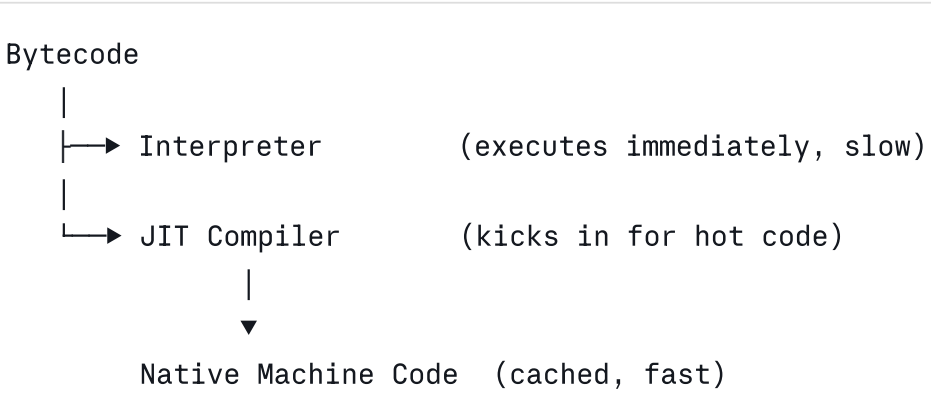


Area	What lives here
Heap	All `new` objects, arrays — shared across threads
Method Area	Class metadata, static fields, constant pool
Stack (per thread)	Stack frames — local variables, operand stack
PC Register	Points to current bytecode instruction for each thread
Native Stack	For native (JNI) method calls

Each **method call** pushes a new **stack frame**; return pops it.

Step 3 — Execution Engine

This is what actually *executes* the bytecode:



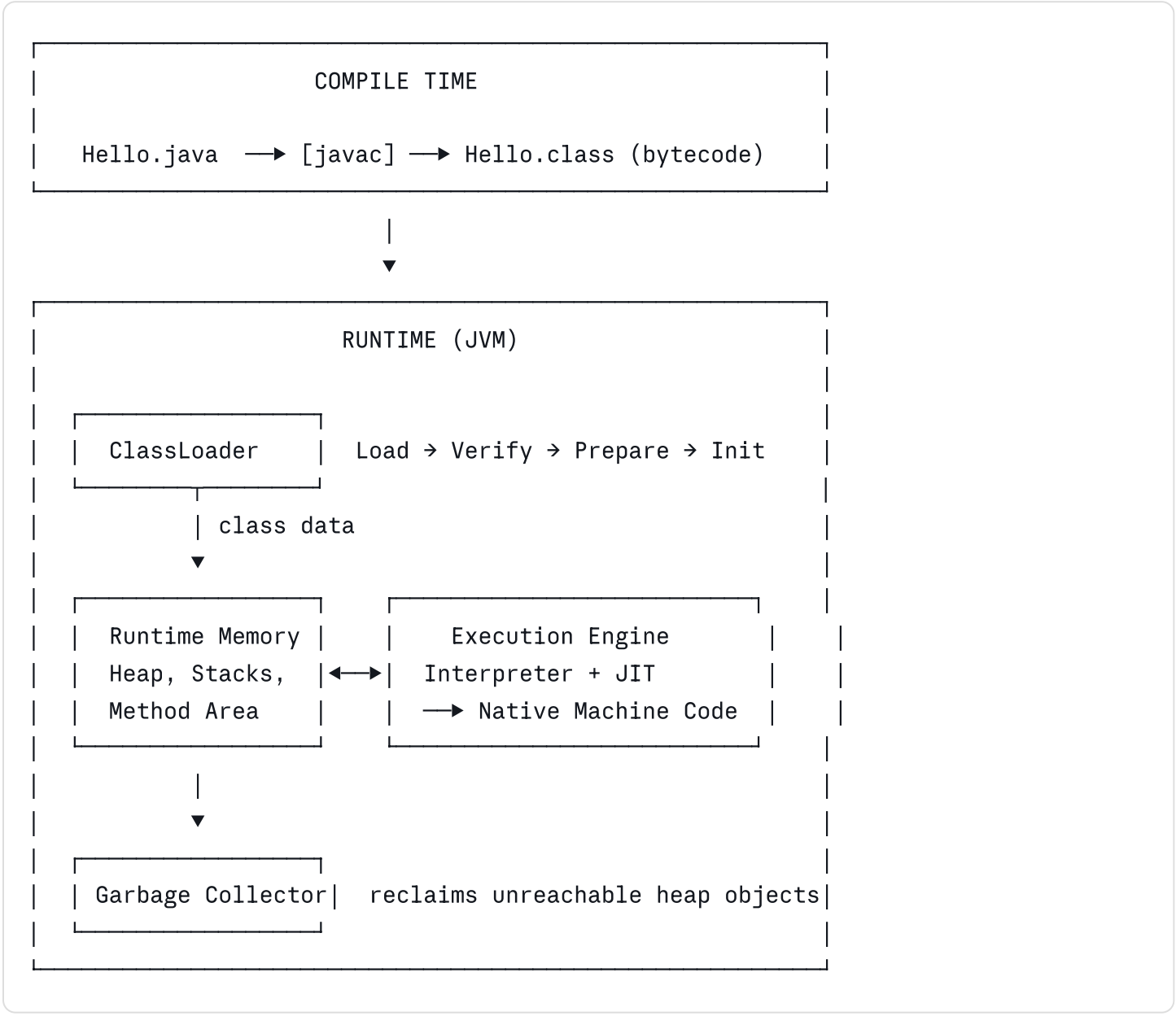
Interpreter runs bytecode line-by-line — fast to start, slow over time.

JIT (Just-In-Time) Compiler monitors which methods run frequently ("hot spots") and compiles them directly to native machine code. This is why Java gets *faster the longer it runs*.

The JIT applies aggressive optimizations:

- **Inlining** — replace method calls with the method body
- **Loop unrolling** — reduce loop overhead
- **Escape analysis** — allocate objects on stack instead of heap when possible
- **Dead code elimination**

Full picture — one diagram



Key takeaways

- `.java` → `.class` happens once via `javac`; the bytecode is OS-independent
- **ClassLoader** loads classes lazily using parent-delegation (security + isolation)
- **Heap** holds objects; **Stack** holds method frames — they serve different purposes
- **JIT** is why Java isn't slow — hot paths become native code at runtime
- **GC** runs in the background reclaiming heap memory; you never call `free()`

Class vs Object explain with example

Class vs Object in Java

Think of it this way:

- A **Class** is the **blueprint** — defined once, describes structure & behavior
- An **Object** is the **instance** — created from the blueprint, lives in memory

---

Class — the blueprint

java

```java
public class Car {

    // Fields — state (each object gets its own copy)
    String brand;
    String color;
    int speed;

    // Constructor — called when object is created
    public Car(String brand, String color) {
        this.brand = brand;
        this.color = color;
        this.speed = 0;
    }

    // Methods — behaviour
    public void accelerate(int amount) {
        speed += amount;
    }

    public void brake(int amount) {
        speed = Math.max(0, speed - amount);
    }

    public void status() {
        System.out.println(brand + " | " + color + " | " + speed + " km/h");
    }
}
```

The class **exists only in code** — no memory is allocated for objects yet.

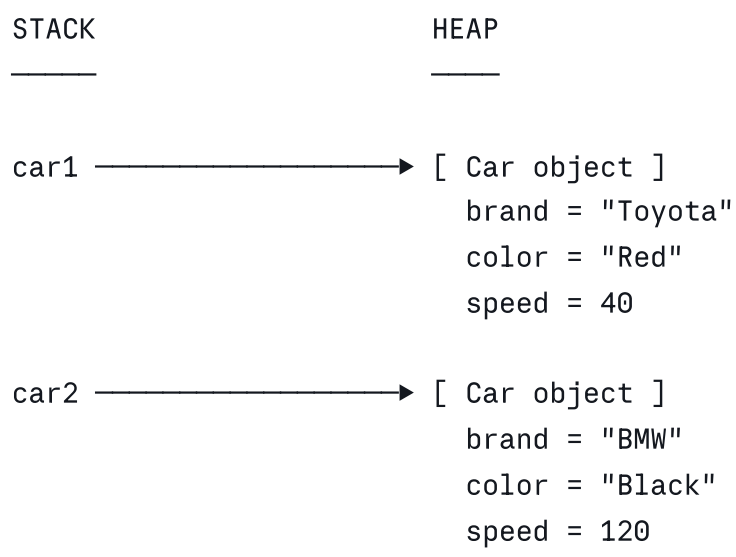
Object — the instance

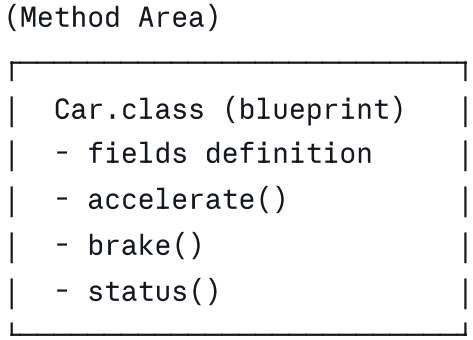
java

```java
public class Main {
    public static void main(String[] args) {

        // Create two objects from the same class
        Car car1 = new Car("Toyota", "Red");
        Car car2 = new Car("BMW", "Black");

        // Each object has its own independent state
        car1.accelerate(60);
        car2.accelerate(120);
        car1.brake(20);

        car1.status();   // Toyota | Red   | 40 km/h
        car2.status();   // BMW    | Black | 120 km/h
    }
}
```

`car1` and `car2` are two **separate objects** in heap memory — same blueprint, independent state.

What happens in memory





```
        ▲           ▲
      car1        car2
  (both share the same method code)
```

- **Variables** (`car1`, `car2`) live on the **stack** — they hold a *reference* (memory address), not the object itself
- **Objects** live on the **heap** — GC manages their lifecycle
- **Method code** lives in the **Method Area** — shared by all instances, not duplicated

---

Class members vs Instance members

java

```java
public class Car {

    // instance field — each object has its OWN copy
    String brand;

    // static field — ONE copy shared by ALL objects
    static int totalCars = 0;

    public Car(String brand) {
        this.brand = brand;
        totalCars++;          // shared counter increments for every new car
    }

    // instance method — works on THIS object's data
    public void show() {
        System.out.println("Car: " + brand);
    }

    // static method — belongs to class, no object needed
    public static void showTotal() {
        System.out.println("Total cars: " + totalCars);
    }
}
```

java

```java
Car c1 = new Car("Tesla");
Car c2 = new Car("Honda");
Car c3 = new Car("Ford");

c1.show();              // Car: Tesla
Car.showTotal();        // Total cars: 3
```

	Instance member	Static member
Belongs to	Each object	The class itself
Memory	Heap (per object)	Method Area (once)
Access via	`objectName.field`	`ClassName.field`
Use case	Per-object state/behaviour	Shared state, utility methods

---

`this` keyword

`this` refers to the **current object** — used to distinguish instance fields from parameters with the same name:

```java
public Car(String brand, String color) {
    this.brand = brand;   // this.brand = instance field
                          //      brand = constructor parameter
    this.color = color;
}
```

Key takeaways

	Class	Object
What	Blueprint / template	Instance in memory
When created	At compile time (`.class` file)	At runtime (`new` keyword)
Where	Method Area	Heap
How many	Once per program	As many as needed
Analogy	Cookie cutter	The actual cookie

> Every object **is-a** instance of its class, but a class **is not** an object — it's the definition. You can create thousands of `Car` objects from one `Car` class, each with its own state, all sharing the same method code.

Constructors (default & parameterized)
Constructor overloading

`this` keyword
Why do we need constructors?

Constructors in Java

A **constructor** is a special method that runs automatically when an object is created with `new`. Its job: **initialize the object's state.**

```java
Car c = new Car("Toyota");
//              ──────────
//              this triggers the constructor
```

Rules:

- Same name as the class
- No return type (not even `void`)
- Called once — at object creation time

Why do we need constructors?

Without a constructor, you'd have to manually set every field after creation — error-prone and easy to forget:

```java
// WITHOUT constructor — dangerous
Car c = new Car();
c.brand = "Toyota";   // what if you forget this?
c.speed = 0;          // object is in invalid state until all fields set
c.fuelLevel = 100;

// WITH constructor — safe, guaranteed initialization
Car c = new Car("Toyota", 100);
// object is fully valid the moment it's created
```

Constructors **enforce object validity** — an object either gets created correctly or not at all.

Default Constructor

A constructor with **no parameters.** If you write no constructor at all, Java silently inserts one for you.

```java
```

```java
public class Car {
    String brand;
    int speed;
    int fuelLevel;

    // Default constructor — no parameters
    public Car() {
        brand    = "Unknown";
        speed    = 0;
        fuelLevel = 100;
        System.out.println("Car created with defaults");
    }
}
```

java

```java
Car c = new Car();
// Output: Car created with defaults
// c.brand = "Unknown", c.speed = 0, c.fuelLevel = 100
```

> **Java's implicit constructor:** If you define *no* constructor, Java auto-generates `public Car() {}` — fields get default values (`null`, `0`, `false`). The moment you write *any* constructor yourself, Java stops generating one.

Parameterized Constructor

A constructor that **accepts arguments** so you can set fields at creation time:

java

```java
public class Car {
    String brand;
    String color;
    int speed;
    int fuelLevel;

    // Parameterized constructor
    public Car(String brand, String color, int fuelLevel) {
        this.brand     = brand;
        this.color     = color;
        this.fuelLevel = fuelLevel;
        this.speed     = 0;         // sensible default for speed
    }
}
```

java

```
Car c1 = new Car("Toyota", "Red",   100);
Car c2 = new Car("BMW",    "Black",  80);
// both objects are fully initialized immediately
```

Constructor Overloading

Just like method overloading — **multiple constructors, different parameter lists**. Java picks the right one based on the arguments you pass.

java

```
public class Car {
    String brand;
    String color;
    int speed;
    int fuelLevel;

    // Constructor 1 — no args, full defaults
    public Car() {
        this.brand     = "Unknown";
        this.color     = "White";
        this.fuelLevel = 100;
        this.speed     = 0;
    }

    // Constructor 2 — brand only
    public Car(String brand) {
        this.brand     = brand;
        this.color     = "White";   // default color
        this.fuelLevel = 100;
        this.speed     = 0;
    }

    // Constructor 3 — brand + color
    public Car(String brand, String color) {
        this.brand     = brand;
        this.color     = color;
        this.fuelLevel = 100;
        this.speed     = 0;
    }

    // Constructor 4 — full control
    public Car(String brand, String color, int fuelLevel) {
        this.brand     = brand;
        this.color     = color;
        this.fuelLevel = fuelLevel;
        this.speed     = 0;
```

```
    }

    public void status() {
        System.out.println(brand + " | " + color + " | fuel: " + fuelLevel + "% | "
    }
}
```

java

```
Car c1 = new Car();
Car c2 = new Car("Toyota");
Car c3 = new Car("BMW", "Black");
Car c4 = new Car("Tesla", "White", 90);

c1.status();  // Unknown | White  | fuel: 100% | 0km/h
c2.status();  // Toyota  | White  | fuel: 100% | 0km/h
c3.status();  // BMW     | Black  | fuel: 100% | 0km/h
c4.status();  // Tesla   | White  | fuel: 90%  | 0km/h
```

`this()` — Constructor Chaining

Repeating default values across constructors is messy. Use `this()` to call one constructor from another — eliminates duplication:

java

```
public class Car {
    String brand;
    String color;
    int fuelLevel;
    int speed;

    // Master constructor — all logic lives HERE
    public Car(String brand, String color, int fuelLevel) {
        this.brand     = brand;
        this.color     = color;
        this.fuelLevel = fuelLevel;
        this.speed     = 0;
        System.out.println("Car initialized: " + brand);
    }

    // Delegates to master constructor ↑
    public Car(String brand, String color) {
        this(brand, color, 100);       // calls 3-arg constructor
    }

    // Delegates to 2-arg constructor ↑
```

```
    public Car(String brand) {
        this(brand, "White");          // calls 2-arg constructor
    }

    // Delegates to 1-arg constructor ↑
    public Car() {
        this("Unknown");               // calls 1-arg constructor
    }
}
```

```
new Car()
    └──► this("Unknown")
             └──► this("Unknown", "White")
                      └──► this("Unknown", "White", 100)
                               └──► actual initialization runs
```

> `this()` **must be the first statement** in the constructor — no exceptions.

`this` keyword — all uses

`this` refers to the **current object instance**. Three main uses:

java

```java
public class Car {
    String brand;
    int speed;

    // USE 1 — distinguish field from parameter (same name)
    public Car(String brand) {
        this.brand = brand;   // this.brand = field, brand = parameter
    }

    // USE 2 — pass current object to another method
    public void register(CarRegistry registry) {
        registry.add(this);   // pass THIS car object to registry
    }

    // USE 3 — call another constructor (shown above)
    public Car() {
        this("Unknown");      // this() = constructor call
    }

    // USE 4 — return current object (enables method chaining)
    public Car accelerate(int amount) {
        this.speed += amount;
        return this;          // return the same object
```

```
    }
}
```

java

```java
// Method chaining thanks to return this
Car c = new Car("Tesla");
c.accelerate(30).accelerate(20).accelerate(10);
//  speed = 60 — each call returns the same object
```

Full mental model

```
new Car("Toyota", "Red", 100)
          |
          ▼
   Java allocates space on HEAP
          |
          ▼
   Matching constructor runs
   this.brand     = "Toyota"
   this.color     = "Red"
   this.fuelLevel = 100
   this.speed     = 0
          |
          ▼
   Reference returned → stored in variable
   Car c = [address → heap object]
```

Summary

Concept	Purpose
Default constructor	Create object with preset/zero values
Parameterized constructor	Create object with caller-supplied values
Constructor overloading	Flexible creation — multiple valid ways to construct
`this.field`	Refer to instance field when name conflicts with parameter
`this()`	Chain constructors — avoid duplicating initialization logic
`return this`	Enable fluent method chaining

> The golden rule: **an object should never exist in an invalid state.** Constructors are your guarantee.

Explain main method in Java and interview questions asked on it

The `main` Method in Java

The `main` method is the **entry point** of every Java application — it's where the JVM starts execution.

```java
public static void main(String[] args) {
    System.out.println("Hello, World!");
}
```

Anatomy — every keyword explained

```java
public static void main(String[] args)
|      |      |    |    └──────────────┘
|      |      |    |     command-line arguments
|      |      |    |     passed as String array
|      |      |    |
|      |      |    └── method name (must be exactly "main")
|      |      |
|      |      └── no return value
|      |          (JVM doesn't expect one back)
|      |
|      └── no object needed to call this
|          JVM calls it directly on the class
|
└── JVM comes from outside the class,
    so it must be accessible to everyone
```

Each keyword is **mandatory** — remove any one of them and the JVM won't recognize it as the entry point.

Why each keyword is required

`public`

The JVM is outside your class. It needs to **access** `main` from anywhere. Making it `public` ensures the JVM can call it without restriction.

```java
// private main — JVM can't see it
private static void main(String[] args) { }  // ❌ won't run
```

`static`

The JVM calls `main` **before creating any object**. Since `static` methods belong to the class (not an instance), the JVM can invoke it without doing `new MyClass()` first.

```java
// non-static main — JVM would need an object first
// but what object? who creates it? chicken-and-egg problem
public void main(String[] args) { }  // ❌ won't run
```

`void`

The JVM doesn't use a return value from `main`. When `main` returns, the program exits. If you want to exit with a status code, use `System.exit(code)` explicitly.

`main`

This is the **contract name** the JVM looks for by default. It's not a keyword — just a convention the JVM relies on.

`String[] args`

An array of `String` arguments passed from the **command line** at launch time. Can be empty but must be present in the signature.

Command-line arguments via `args`

```java
public class Greet {
    public static void main(String[] args) {
        if (args.length == 0) {
            System.out.println("No name provided.");
        } else {
            System.out.println("Hello, " + args[0] + "!");
        }
    }
}
```

```bash
```

```
javac Greet.java
java Greet Alice        # Hello, Alice!
java Greet              # No name provided.
java Greet Alice Bob    # Hello, Alice!  (args[1] = "Bob" is ignored here)
```

`args` is just a regular `String[]` — you can loop it, parse it, convert types:

java

```java
public static void main(String[] args) {
    int sum = 0;
    for (String arg : args) {
        sum += Integer.parseInt(arg);   // convert String → int
    }
    System.out.println("Sum: " + sum);
}
```

bash

```
java Sum 10 20 30    # Sum: 60
```

Valid `main` signatures Java accepts

java

```java
// All of these are valid — JVM recognizes them
public static void main(String[] args)      // classic
public static void main(String args[])      // array syntax variant
static public void main(String[] args)      // modifier order doesn't matter
public static final void main(String[] args)    // final is allowed
public static synchronized void main(String[] args) // synchronized is allowed
public static void main(String... args)     // varargs — valid since Java 5
```

java

```java
// These will NOT work as entry points
public void main(String[] args)             // ❌ missing static
static void main(String[] args)             // ❌ missing public
public static int main(String[] args)       // ❌ wrong return type
public static void Main(String[] args)      // ❌ wrong name (case-sensitive)
public static void main()                   // ❌ missing args parameter
```

Can `main` be overloaded?

Yes — but the JVM only calls `main(String[] args)`:

java

```java
public class Demo {

    // JVM calls THIS one
    public static void main(String[] args) {
        System.out.println("Entry point");
        main(42);           // you can call overloaded version manually
    }

    // Overloaded — valid Java, but JVM won't call this directly
    public static void main(int number) {
        System.out.println("Overloaded main: " + number);
    }
}
```

```
Entry point
Overloaded main: 42
```

Can `main` be overridden?

No — `static` methods belong to the class, not an instance. They are **hidden**, not overridden, in subclasses:

java

```java
class Parent {
    public static void main(String[] args) {
        System.out.println("Parent main");
    }
}

class Child extends Parent {
    public static void main(String[] args) {
        System.out.println("Child main");     // this is method hiding, not overridi
    }
}
```

bash

```bash
java Parent    # Parent main
java Child     # Child main
```

Each class has its own independent `main` — the JVM picks based on which class you run.

`main` inside flow — what JVM does step by step

```
java MyApp
     |
     ▼
JVM starts up
     |
     ▼
Bootstrap ClassLoader loads java.lang.*
     |
     ▼
Application ClassLoader loads MyApp.class
     |
     ▼
JVM looks for: public static void main(String[])
     |
     ├── not found → error: Main method not found
     |
     └── found ──►  new thread created ("main" thread)
                          |
                          ▼
                    Stack frame for main() pushed
                          |
                          ▼
                    Your code runs
                          |
                          ▼
                    main() returns
                          |
                          ▼
                    JVM checks for non-daemon threads
                          |
                    none running → JVM exits
```

Interview Questions on `main`

Q1. Can you run a Java program without a `main` method?

In modern Java — **no.** But historically:

- Before Java 7 — you could run code via `static` blocks (JVM ran them then crashed with no main found, but output was visible)
- Java 7+ — JVM explicitly validates `main` exists before running static blocks

```
java
```

```
// This used to print then crash pre-Java 7
public class Trick {
    static {
        System.out.println("Static block ran!");
        System.exit(0);    // prevented the crash
    }
}
// Java 7+: Error: Main method not found — static block never runs
```

Q2. Why is `main` `static`?

Because the JVM needs to call it **without creating an object first**. There's no mechanism to tell the JVM which constructor to use or what arguments to pass — so `main` must be reachable at the class level via `static`.

Q3. What happens if you remove `String[] args`?

java

```
public static void main() { }   // ❌
```

The JVM looks for exactly `main(String[])`. This signature doesn't match — `Main method not found` error at runtime even though it compiles fine.

Q4. Can `main` be `final`?

java

```
public static final void main(String[] args) { }   // ✅ runs fine
```

Yes — `final` just prevents subclasses from hiding it (though since it's static, hiding was already not polymorphic). JVM doesn't care.

Q5. Can `main` throw exceptions?

java

```
public static void main(String[] args) throws Exception {   // ✅
    throw new Exception("Crash!");
}
```

Yes — if an uncaught exception escapes `main`, the JVM prints the stack trace and exits.

Q6. Can a class have multiple `main` methods?

Yes — through **overloading**. But the JVM only uses `main(String[] args)` as the entry point. Other `main` variants are just regular static methods.

Q7. Can `main` be `private` or `protected`?

```java
private static void main(String[] args) { }    // compiles but JVM can't call it
```

It compiles but the JVM requires `public` access to invoke it. You'll get `Main method not found` at runtime.

Q8. Is `main` a keyword in Java?

No. `main` is just an identifier — a regular method name that the JVM happens to look for by convention. You could name a variable `main`, or have a class method `main` that does something entirely different (though that's poor practice).

Q9. What is the return type of `main` and can you change it?

Return type must be `void`. The JVM doesn't collect any return value from `main`. If you want to signal an exit code to the OS, use:

```java
System.exit(0);    // success
System.exit(1);    // failure / abnormal exit
```

Q10. Can you call `main` explicitly from within a program?

Yes — it's just a static method. You can call it like any other:

```java
public class Demo {
    public static void main(String[] args) {
        System.out.println("main called");
        if (args.length == 0) {
            main(new String[]{"recursive"});   // call main explicitly
        }
    }
}
```

```
main called
main called
```

This won't cause infinite recursion here because of the `args.length` check — but it shows `main` is just a regular callable method.

Quick reference card

Aspect	Answer
Access modifier	Must be `public`
Static?	Must be `static`
Return type	Must be `void`
Method name	Must be exactly `main`
Parameter	Must be `String[]` or `String...`
Can be overloaded?	Yes
Can be overridden?	No (static → method hiding)
Can be `final`?	Yes
Can throw exceptions?	Yes
Is `main` a keyword?	No — just a convention

Access Modifiers
Packages & Imports

Access Modifiers & Packages in Java

Access Modifiers

Access modifiers control **who can see and use** a class, field, method, or constructor.

Java has four levels:

```
Most restrictive ◄──────────────────────────► Least restrictive
   private    →    default    →    protected    →    public
```

The four modifiers explained

`private` — class only

java

```java
public class BankAccount {
    private double balance;      // only THIS class can touch this
    private String pin;

    public BankAccount(double initialBalance, String pin) {
        this.balance = initialBalance;
        this.pin = pin;
    }

    private boolean validatePin(String input) {   // internal helper
        return this.pin.equals(input);
    }

    public void withdraw(String pin, double amount) {
        if (validatePin(pin)) {          // private method used internally
            balance -= amount;
        } else {
            System.out.println("Wrong PIN!");
        }
    }

    public double getBalance() {         // controlled access via public method
        return balance;
    }
}
```

java

```java
BankAccount acc = new BankAccount(5000, "1234");
acc.balance;          // ❌ compile error — private
acc.validatePin();    // ❌ compile error — private
acc.getBalance();     // ✅ works — public method exposes it safely
```

`default` (package-private) — no keyword written

When you write **no modifier**, access is limited to classes in the **same package**:

java

```java
// file: com/bank/Transaction.java
package com.bank;

class Transaction {              // no modifier = package-private
    double amount;               // package-private field
    String type;

    void process() {             // package-private method
        System.out.println("Processing " + type + ": " + amount);
    }
}
```

java

```java
// file: com/bank/BankAccount.java — SAME package ✅
package com.bank;

public class BankAccount {
    void makeTransaction() {
        Transaction t = new Transaction();   // ✅ same package
        t.process();                         // ✅ accessible
    }
}
```

java

```java
// file: com/app/Main.java — DIFFERENT package ❌
package com.app;
import com.bank.Transaction;     // ❌ compile error — not public
```

`protected` — package + subclasses

java

```java
// file: com/animals/Animal.java
package com.animals;

public class Animal {
    protected String name;
    protected int age;

    protected void breathe() {
        System.out.println(name + " is breathing");
    }
```

```java
    private void heartbeat() {       // only Animal itself
        System.out.println("heartbeat");
    }
}
```

java

```java
// file: com/animals/Dog.java — same package ✅
package com.animals;

public class Dog extends Animal {
    public void bark() {
        System.out.println(name + " barks!");     // ✅ protected field
        breathe();                                 // ✅ protected method
    }
}
```

java

```java
// file: com/pets/GuideDog.java — different package, but subclass ✅
package com.pets;
import com.animals.Animal;

public class GuideDog extends Animal {
    public void guide() {
        System.out.println(name + " is guiding");  // ✅ inherited protected
        breathe();                                  // ✅ inherited protected
    }
}
```

java

```java
// file: com/app/Main.java — different package, NOT a subclass ❌
package com.app;
import com.animals.Animal;

public class Main {
    public static void main(String[] args) {
        Animal a = new Animal();
        a.name = "Cat";      // ❌ protected — not a subclass here
        a.breathe();         // ❌ protected — not a subclass here
    }
}
```

`public` — everywhere

java

```java
package com.utils;

public class MathUtils {
    public static int add(int a, int b) {    // accessible from anywhere
        return a + b;
    }
}
```

java

```java
// from any package, any class
import com.utils.MathUtils;
int result = MathUtils.add(3, 5);    // ✅ works anywhere
```

Access modifier comparison table

Modifier	Same Class	Same Package	Subclass (diff pkg)	Everywhere
`private`	✅	❌	❌	❌
`default`	✅	✅	❌	❌
`protected`	✅	✅	✅	❌
`public`	✅	✅	✅	✅

What can each modifier apply to?

Modifier	Class (top-level)	Field	Method	Constructor
`private`	❌	✅	✅	✅
`default`	✅	✅	✅	✅
`protected`	❌	✅	✅	✅
`public`	✅	✅	✅	✅

Top-level classes can only be `public` or package-private — never `private` or `protected`.

Packages

A **package** is a namespace that groups related classes together — like folders for your `.java` files.

```
src/
└── com/
    └── bank/
        ├── BankAccount.java     → package com.bank
        ├── Transaction.java     → package com.bank
        └── utils/
            └── CurrencyUtil.java → package com.bank.utils
```

Declaring a package

Must be the **first line** of every `.java` file:

```java
package com.bank;           // this file belongs to com.bank package

public class BankAccount {
    // ...
}
```

If you write no `package` declaration, the class goes into the **default (unnamed) package** — fine for quick experiments, bad practice in real projects.

Why use packages?

```
Without packages                With packages
────────────────                ─────────────
BankAccount.java                com.bank.BankAccount
Transaction.java                com.bank.Transaction
Date.java              →        com.bank.utils.Date
List.java                       com.collections.List
String.java                     java.lang.String

Name clashes everywhere!        No clash — full name is unique
```

Packages solve three problems: **organization, name collision, access control.**

Importing classes

java

```java
package com.app;

// Import a specific class
import com.bank.BankAccount;

// Import everything in a package (wildcard)
import com.bank.*;

// Static import — use static members without class name
import static java.lang.Math.sqrt;
import static java.lang.Math.PI;

public class Main {
    public static void main(String[] args) {
        BankAccount acc = new BankAccount(5000, "1234");   // no prefix needed

        double result = sqrt(16);    // instead of Math.sqrt(16)
        double area = PI * 5 * 5;    // instead of Math.PI
    }
}
```

Import rules & behavior

java

```java
// ✅ specific import — preferred, clear what's being used
import java.util.ArrayList;
import java.util.HashMap;

// ✅ wildcard — imports ALL classes in java.util (not subpackages)
import java.util.*;

// ❌ wildcard does NOT import subpackages
import java.util.*;
// java.util.concurrent.CopyOnWriteArrayList still needs explicit import
import java.util.concurrent.CopyOnWriteArrayList;
```

`java.lang` **is always auto-imported** — that's why you can use `String`, `System`, `Math`, `Integer` without importing:

java

```java
// These are implicitly available everywhere
String s = "hello";          // java.lang.String
```

```
System.out.println(s);        // java.lang.System
Math.abs(-5);                 // java.lang.Math
```

Name collision — how to resolve

java

```
import java.util.Date;
import java.sql.Date;         // ❌ compile error — ambiguous
```

Solution — use **fully qualified name** for one of them:

java

```
import java.util.Date;        // import one

public class Main {
    public static void main(String[] args) {
        Date utilDate = new Date();                    // java.util.Date
        java.sql.Date sqlDate = new java.sql.Date(0); // fully qualified
    }
}
```

Package naming conventions

```
com.companyname.projectname.module

com.google.gson
com.amazon.s3.client
org.apache.commons.lang
java.util.concurrent

// Your projects
com.yourname.bankapp.accounts
com.yourname.bankapp.utils
```

Always **lowercase**, reversed domain name first — guarantees global uniqueness.

Full working example

```
src/
├── com/bank/
|   ├── BankAccount.java
```

```
|   └── utils/
|       └── CurrencyUtil.java
└── com/app/
    └── Main.java
```

java

```java
// com/bank/utils/CurrencyUtil.java
package com.bank.utils;

public class CurrencyUtil {
    public static String format(double amount) {
        return String.format("₹%.2f", amount);
    }
}
```

java

```java
// com/bank/BankAccount.java
package com.bank;

import com.bank.utils.CurrencyUtil;   // import from subpackage

public class BankAccount {
    private double balance;

    public BankAccount(double balance) {
        this.balance = balance;
    }

    public void showBalance() {
        System.out.println("Balance: " + CurrencyUtil.format(balance));
    }
}
```

java

```java
// com/app/Main.java
package com.app;

import com.bank.BankAccount;          // import from another package

public class Main {
    public static void main(String[] args) {
        BankAccount acc = new BankAccount(5000.50);
        acc.showBalance();    // Balance: ₹5000.50
    }
}
```

Summary

```
Access Modifiers — control visibility
──────────────────────────────────────

private    → class only            (encapsulation)
default    → package only          (internal collaboration)
protected  → package + subclasses  (inheritance)
public     → everywhere            (API surface)

Packages — control organization
────────────────────────────────

package com.bank;          → declare membership
import com.bank.Account;   → use a specific class
import com.bank.*;         → use all classes in package
import static Math.sqrt;   → use static members directly
java.lang.*                → always auto-imported
```

> **Rule of thumb:** keep everything as restrictive as possible. Start with `private`, loosen only when needed. This is the foundation of **encapsulation** — a core OOP principle.

Data types — primitive (`int`, `char`, `boolean`, `double`, etc.)

Data Types in Java

Java is **statically typed** — every variable must have a declared type known at compile time. There are two categories:

```
Java Data Types
├── Primitive    — 8 types, store actual value, live on stack
└── Reference    — Objects, arrays, Strings — store address, live on heap
```

The 8 Primitive Types

Type	Size	Default	Range	Example literal
`byte`	1 byte	0	-128 to 127	`byte b = 100;`
`short`	2 bytes	0	-32,768 to 32,767	`short s = 3000;`

Type	Size	Default	Range	Example literal
`int`	4 bytes	0	-2.1B to 2.1B	`int i = 42;`
`long`	8 bytes	0L	-9.2Q to 9.2Q	`long l = 99L;`
`float`	4 bytes	0.0f	~7 decimal digits	`float f = 3.14f;`
`double`	8 bytes	0.0d	~15 decimal digits	`double d = 3.14;`
`char`	2 bytes	`\u0000`	0 to 65,535 (Unicode)	`char c = 'A';`
`boolean`	~1 bit*	`false`	`true` / `false`	`boolean flag = true;`

*JVM-dependent, not precisely defined in spec.

Integer types

java

```java
byte   b = 120;              // tiny — good for raw binary data
short  s = 30_000;           // rarely used
int    i = 2_147_483_647;    // most common — default for integers
long   l = 9_876_543_210L;   // needs L suffix — large numbers

// Underscores in literals (Java 7+) — improves readability
int million   = 1_000_000;
long cardNum  = 1234_5678_9012_3456L;

// Different bases
int decimal = 255;
int hex     = 0xFF;       // 255 — prefix 0x
int octal   = 0377;       // 255 — prefix 0
int binary  = 0b11111111; // 255 — prefix 0b (Java 7+)

System.out.println(decimal == hex);    // true — all are 255
```

Overflow — wraps around silently:

java

```java
int max = Integer.MAX_VALUE;     // 2147483647
System.out.println(max + 1);     // -2147483648  ← wraps around!

// Use long for big numbers
long bigNum = (long) max + 1;    // 2147483648 ✅
```

Floating point types

```java
float  f = 3.14f;        // needs f suffix, ~7 sig digits
double d = 3.14159265;   // default for decimals, ~15 sig digits

// Special values
double inf     = 1.0 / 0.0;          // Infinity
double negInf  = -1.0 / 0.0;         // -Infinity
double notNum  = 0.0 / 0.0;          // NaN (Not a Number)

System.out.println(inf);       // Infinity
System.out.println(notNum);    // NaN
System.out.println(Double.isNaN(notNum));      // true
System.out.println(Double.isInfinite(inf));    // true
```

⚠️ Floating point precision trap:

```java
double a = 0.1 + 0.2;
System.out.println(a);              // 0.30000000000000004  ← NOT 0.3!
System.out.println(a == 0.3);       // false  ← never compare with ==

// Fix 1 — round for display
System.out.printf("%.2f%n", a);     // 0.30

// Fix 2 — use BigDecimal for exact arithmetic (money, etc.)
import java.math.BigDecimal;
BigDecimal x = new BigDecimal("0.1");
BigDecimal y = new BigDecimal("0.2");
System.out.println(x.add(y));       // 0.3  ✅
```

> **Rule:** Never use `float` / `double` for money. Use `BigDecimal`.

`char` type

Stores a **single Unicode character** as a 16-bit unsigned integer:

```java
char letter  = 'A';          // single quotes — not double!
char digit   = '9';
char space   = ' ';
char unicode = '\u0041';     // 'A' — Unicode escape
char newline = '\n';         // escape sequence
```

```
// char is numeric underneath
char c = 'A';
System.out.println(c);         // A
System.out.println((int) c);   // 65  ← ASCII/Unicode value

// Arithmetic on chars
char next = (char) (c + 1);
System.out.println(next);      // B

// Iterating alphabet
for (char ch = 'a'; ch <= 'z'; ch++) {
    System.out.print(ch + " ");
}
// a b c d e f g h i j k l m n o p q r s t u v w x y z
```

Common escape sequences:

Sequence	Meaning
`'\n'`	newline
`'\t'`	tab
`'\\'`	backslash
`'\''`	single quote
`'\"'`	double quote
`'\u0041'`	Unicode (A)

`boolean` type

Only two values — `true` or `false`:

java

```
boolean isLoggedIn  = true;
boolean hasBalance  = false;

// Result of comparisons is boolean
boolean isAdult     = (age >= 18);
boolean isValid     = (balance > 0 && !isBlocked);

// Used in control flow
if (isLoggedIn && hasBalance) {
    System.out.println("Allow transaction");
}
```

```
// boolean cannot be cast to int (unlike C/C++)
int x = (int) true;    // ❌ compile error in Java
```

Type ranges — how to remember

```
byte   → 1 byte  → -2⁷   to 2⁷-1   →    -128 to 127
short  → 2 bytes → -2¹⁵  to 2¹⁵-1  → -32,768 to 32,767
int    → 4 bytes → -2³¹  to 2³¹-1  →      ~-2B to 2B
long   → 8 bytes → -2⁶³  to 2⁶³-1  →      ~-9Q to 9Q
```

Access them via constants:

java

```
System.out.println(Byte.MAX_VALUE);        //  127
System.out.println(Byte.MIN_VALUE);        // -128
System.out.println(Integer.MAX_VALUE);     //  2147483647
System.out.println(Long.MAX_VALUE);        //  9223372036854775807
System.out.println(Double.MAX_VALUE);      //  1.7976931348623157E308
```

Type casting

Two kinds — **widening** (automatic) and **narrowing** (manual):

```
byte → short → int → long → float → double
                                ↑
                          char ─┘
```

Widening — automatic, no data loss

java

```
int    i = 100;
long   l = i;        // ✅ int → long, automatic
double d = l;        // ✅ long → double, automatic
float  f = i;        // ✅ int → float, automatic

// char to int — widening
char c = 'A';
int  n = c;          // ✅ n = 65
```

Narrowing — manual cast required, may lose data

java

```
double d = 9.99;
int    i = (int) d;        // ✅ i = 9  — decimal part lost!

long   l = 1234567890123L;
int    x = (int) l;        // ✅ but value may be wrong — overflow risk

int    n = 65;
char   c = (char) n;       // ✅ c = 'A'

double price = 9.99;
float  f     = (float) price;   // some precision lost
```

Tricky casting cases

java

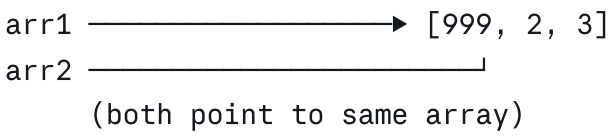
```
// int + int = int, even if stored in long
long result = 1_000_000 * 1_000_000;    // ❌ overflows int first!
long result = 1_000_000L * 1_000_000;   // ✅ forces long arithmetic

// byte arithmetic promotes to int automatically
byte a = 10, b = 20;
byte c = a + b;             // ❌ a+b produces int, needs cast
byte c = (byte)(a + b);     // ✅
```

Primitives vs Reference types

java

```
// Primitive — stores the VALUE directly
int a = 10;
int b = a;      // b gets a COPY of the value
b = 99;
System.out.println(a);   // 10 — a is unaffected

// Reference — stores an ADDRESS to the object
int[] arr1 = {1, 2, 3};
int[] arr2 = arr1;          // arr2 points to SAME array
arr2[0] = 999;
System.out.println(arr1[0]); // 999 — arr1 IS affected!
```

```
STACK                   HEAP
_____                   ____


a = 10                  (nothing — value is directly on stack)
b = 99
```



Wrapper classes

Every primitive has an **object wrapper** in `java.lang`:

Primitive	Wrapper
`int`	`Integer`
`long`	`Long`
`double`	`Double`
`float`	`Float`
`char`	`Character`
`boolean`	`Boolean`
`byte`	`Byte`
`short`	`Short`

Used when you need an object — collections, generics, nullability:

java

```java
// Autoboxing — primitive automatically wrapped into object
Integer obj = 42;           // int → Integer (automatic)

// Unboxing — object automatically unwrapped to primitive
int val = obj;              // Integer → int (automatic)

// Needed for collections (generics require objects, not primitives)
List<Integer> list = new ArrayList<>();
list.add(10);               // autoboxing: 10 → Integer(10)
int x = list.get(0);        // unboxing: Integer(10) → 10

// Useful methods on wrappers
Integer.parseInt("123");          // String → int
Integer.toBinaryString(255);      // "11111111"
Integer.toHexString(255);         // "ff"
Double.parseDouble("3.14");       // String → double
Character.isDigit('5');           // true
```

```
Character.isLetter('A');          // true
Character.toUpperCase('a');       // 'A'
```

Autoboxing trap — `==` vs `.equals()`:

java

```
Integer x = 127;
Integer y = 127;
System.out.println(x == y);       // true  (cached range: -128 to 127)

Integer p = 128;
Integer q = 128;
System.out.println(p == q);       // false (outside cache — different objects)
System.out.println(p.equals(q));  // true  ← always use equals() for wrappers
```

Summary

```
Integer types  →  byte  short  int(default)  long(L suffix)
Decimal types  →  float(f suffix)  double(default)
Text/char      →  char (single quotes, Unicode)
Logic          →  boolean (true/false only)

Widening  →  automatic  (small → large, no data loss)
Narrowing →  manual cast (large → small, possible data loss)

Primitives  →  value stored on stack, copied by value
Wrappers    →  object on heap, needed for collections/generics
BigDecimal  →  use for money/exact decimal arithmetic
```

Why do we need Wrapper Classes?
Difference between `int` and `Integer`?
Why use BigDecimal instead of double?

Wrapper Classes, int vs Integer, and BigDecimal

Why do we need Wrapper Classes?

Java's primitives (`int`, `double`, etc.) are **not objects** — they're raw values. But many parts of Java **require objects**. Wrapper classes bridge that gap.

Reason 1 — Collections only work with objects

java

```java
// ❌ primitives not allowed in generics
List<int>     list = new ArrayList<>();   // compile error
Map<int, int> map  = new HashMap<>();     // compile error

// ✅ wrapper classes work
List<Integer>        list = new ArrayList<>();
Map<String, Integer> map  = new HashMap<>();

list.add(10);     // int → Integer (autoboxing, automatic)
int x = list.get(0);  // Integer → int (unboxing, automatic)
```

Reason 2 — Nullability

A primitive **cannot be null** — it always has a value. A wrapper **can be null**, which is useful when absence of a value is meaningful:

java

```java
int     age = null;      // ❌ compile error — primitives can't be null
Integer age = null;      // ✅ means "age is unknown / not provided"

// Real-world use — database fields are often nullable
Integer salary = fetchFromDatabase("salary");   // might return null
if (salary != null) {
    System.out.println("Salary: " + salary);
} else {
    System.out.println("Salary not on record");
}
```

Reason 3 — Utility methods

Wrapper classes come packed with useful static methods:

java

```java
// Parsing strings → primitives
int    i = Integer.parseInt("123");
double d = Double.parseDouble("3.14");
long   l = Long.parseLong("9876543210");
boolean b = Boolean.parseBoolean("true");

// Number conversions
```

```java
String binary = Integer.toBinaryString(255);    // "11111111"
String hex    = Integer.toHexString(255);        // "ff"
String octal  = Integer.toOctalString(255);      // "377"

// Bounds
System.out.println(Integer.MAX_VALUE);    // 2147483647
System.out.println(Double.MIN_VALUE);     // 4.9E-324

// Comparison utilities
Integer.compare(5, 10);          // -1 (5 < 10)
Integer.max(5, 10);              // 10
Integer.sum(5, 10);              // 15
Integer.min(5, 10);              // 5

// Character utilities
Character.isDigit('5');          // true
Character.isLetter('A');         // true
Character.isWhitespace(' ');     // true
Character.toUpperCase('a');      // 'A'
Character.toLowerCase('Z');      // 'z'
```

Reason 4 — Generics & type parameters

java

```java
// Generic method needs a type — must be an object
public <T extends Comparable<T>> T findMax(T a, T b) {
    return a.compareTo(b) > 0 ? a : b;
}

findMax(10, 20);          // ✅ Integer (autoboxed)
findMax(3.14, 2.71);      // ✅ Double  (autoboxed)
findMax("cat", "dog");    // ✅ String
```

Reason 5 — Reflection & Object-based APIs

java

```java
// Reflection requires objects
Object obj = 42;                          // autoboxed to Integer
System.out.println(obj.getClass());       // class java.lang.Integer

// Optional — wraps nullable values cleanly
Optional<Integer> maybeAge = Optional.of(25);
Optional<Integer> noAge    = Optional.empty();
```

`int` vs `Integer` — full comparison

	`int`	`Integer`
Type	Primitive	Object (reference)
Memory	4 bytes on stack	~16 bytes on heap + reference
Default value	`0`	`null`
Can be null?	❌ No	✅ Yes
In collections?	❌ No	✅ Yes
Has methods?	❌ No	✅ Yes
Comparison	`==` always works	`==` compares reference
Performance	Faster	Slower (object overhead)

The `==` trap with Integer

java

```
// int — == always compares values
int a = 1000;
int b = 1000;
System.out.println(a == b);       // true ✅

// Integer — == compares object references (addresses)
Integer x = 1000;
Integer y = 1000;
System.out.println(x == y);       // false ❌ — different objects on heap
System.out.println(x.equals(y));  // true  ✅ — always use equals()
```

Why does this work for small numbers though?

java

```
Integer p = 100;
Integer q = 100;
System.out.println(p == q);       // true  — wait, what?
```

Java caches `Integer` objects in the range **-128 to 127** (the `IntegerCache`). Within this range, `new Integer` is not created — the same cached object is returned. Outside this range, new objects are allocated.

java

```java
// Inside cache (-128 to 127) — same object returned
Integer a = 127;  Integer b = 127;
System.out.println(a == b);    // true  (same cached object)

// Outside cache — new objects
Integer c = 128;  Integer d = 128;
System.out.println(c == d);    // false (different objects)
System.out.println(c.equals(d)); // true ✅ — always use equals()
```

> **Rule:** Always use `.equals()` to compare `Integer` objects. Never `==`.

Autoboxing & Unboxing — the automatic conversion

java

```java
// Autoboxing — int → Integer (compiler inserts Integer.valueOf())
Integer obj = 42;
// compiler rewrites to: Integer obj = Integer.valueOf(42);

// Unboxing — Integer → int (compiler inserts .intValue())
int val = obj;
// compiler rewrites to: int val = obj.intValue();
```

Autoboxing trap — NullPointerException:

java

```java
Integer score = null;    // valid — Integer can be null
int    total  = score;   // ❌ NullPointerException!
                         // unboxing null calls null.intValue() → crash

// Always null-check before unboxing
if (score != null) {
    int total = score;
}
```

Autoboxing trap — performance in loops:

java

```java
// ❌ BAD — creates millions of Integer objects on heap
Long sum = 0L;
for (long i = 0; i < 1_000_000; i++) {
    sum += i;    // unbox sum, add i, rebox result — every iteration!
}
```

```
// ✅ GOOD — stays on stack, no object creation
long sum = 0L;
for (long i = 0; i < 1_000_000; i++) {
    sum += i;
}
```

Why use `BigDecimal` instead of `double`?

Because `double` uses **binary floating-point** (IEEE 754) — it cannot represent most decimal fractions exactly.

The core problem

java

```
System.out.println(0.1 + 0.2);          // 0.30000000000000004
System.out.println(1.03 - 0.42);        // 0.6099999999999999
System.out.println(0.1 * 3);            // 0.30000000000000004
System.out.println(4.0 / 3.0);          // 1.3333333333333333

// Money calculation with double — DANGEROUS
double price    = 19.99;
double tax      = 0.05;
double total    = price + (price * tax);
System.out.println(total);              // 20.9895 — should be 20.989500...
```

Why does this happen? `0.1` in binary is `0.000110011001100...` (repeating) — just like `1/3` in decimal is `0.333...`. It can never be stored exactly in a finite number of bits.

BigDecimal — exact decimal arithmetic

java

```
import java.math.BigDecimal;
import java.math.RoundingMode;

// Always use String constructor — not double!
BigDecimal a = new BigDecimal("0.1");
BigDecimal b = new BigDecimal("0.2");

System.out.println(a.add(b));           // 0.3  ✅ exact

// ❌ double constructor still has imprecision
```

```java
BigDecimal bad = new BigDecimal(0.1);
System.out.println(bad);   // 0.1000000000000000055511151231257827021181583404541015
```

BigDecimal operations

java

```java
BigDecimal price    = new BigDecimal("19.99");
BigDecimal taxRate  = new BigDecimal("0.05");
BigDecimal quantity = new BigDecimal("3");

// Arithmetic
BigDecimal tax      = price.multiply(taxRate);             // 0.9995
BigDecimal total    = price.add(tax);                      // 20.9895
BigDecimal subtotal = price.multiply(quantity);            // 59.97
BigDecimal discount = subtotal.subtract(new BigDecimal("5.00")); // 54.97
BigDecimal perItem  = subtotal.divide(quantity);           // 19.99

// Rounding — essential for money
BigDecimal rounded = total.setScale(2, RoundingMode.HALF_UP);
System.out.println(rounded);    // 20.99

// Comparison — use compareTo(), NOT equals()!
BigDecimal x = new BigDecimal("2.0");
BigDecimal y = new BigDecimal("2.00");
System.out.println(x.equals(y));        // false (scale differs!)
System.out.println(x.compareTo(y));     // 0  ✅ (numerically equal)
```

RoundingMode options

java

```java
BigDecimal n = new BigDecimal("2.345");

n.setScale(2, RoundingMode.HALF_UP);    // 2.35  — standard rounding
n.setScale(2, RoundingMode.HALF_DOWN);  // 2.34
n.setScale(2, RoundingMode.CEILING);    // 2.35  — always up
n.setScale(2, RoundingMode.FLOOR);      // 2.34  — always down
n.setScale(2, RoundingMode.UP);         // 2.35  — away from zero
n.setScale(2, RoundingMode.DOWN);       // 2.34  — toward zero
n.setScale(2, RoundingMode.HALF_EVEN);  // 2.34  — banker's rounding
```

Real-world money example

java

```java
import java.math.BigDecimal;
import java.math.RoundingMode;

public class Invoice {

    static BigDecimal applyTax(BigDecimal amount, BigDecimal taxRate) {
        return amount
            .multiply(taxRate)
            .setScale(2, RoundingMode.HALF_UP);
    }

    public static void main(String[] args) {
        BigDecimal price    = new BigDecimal("49.99");
        BigDecimal taxRate  = new BigDecimal("0.18");   // 18% GST
        BigDecimal quantity = new BigDecimal("4");

        BigDecimal subtotal = price.multiply(quantity);                 // 199.96
        BigDecimal tax      = applyTax(subtotal, taxRate);              //  35.99
        BigDecimal total    = subtotal.add(tax);                        // 235.95

        System.out.println("Subtotal : ₹" + subtotal);   // ₹199.96
        System.out.println("GST 18%  : ₹" + tax);        // ₹35.99
        System.out.println("Total    : ₹" + total);      // ₹235.95
    }
}
```

When to use what

```
int       → counting, indexing, general integers — default choice
Integer   → collections, nullability, parsing, generics
long      → large integers (timestamps, IDs, file sizes)
double    → scientific calculations where tiny errors are acceptable
float     → memory-sensitive bulk data (graphics, sensors)
BigDecimal→ money, tax, financial calculations — anywhere exactness matters
```

Summary

```
Wrapper classes needed for:
  ├── Collections / Generics   (List<Integer>, Map<String, Double>)
  ├── Nullability              (Integer can be null, int cannot)
  ├── Utility methods          (parseInt, toBinaryString, compare)
  └── Object-based APIs        (Reflection, Optional, streams)
```

```
int vs Integer:
  ├── int     → fast, stack, no null, no methods
  ├── Integer → object, heap, nullable, has methods
  └── ==  on Integer compares address — always use .equals()

double vs BigDecimal:
  ├── double     → fast, approximate, fine for science/display
  ├── BigDecimal → exact, slow, mandatory for money
  └── Always construct BigDecimal from String, never from double
```

Variables — local, instance, static
Explain with examples along with questions asked in interview

Variables in Java — Local, Instance, Static

A variable is a **named memory location** that holds a value. Where you declare it determines its **scope, lifetime, default value, and memory location.**

Three types at a glance

```
class BankAccount {

    static int totalAccounts = 0;    // ← STATIC variable
    String owner;                    // ← INSTANCE variable
    double balance;                  // ← INSTANCE variable

    void deposit(double amount) {
        double newBalance = balance + amount;  // ← LOCAL variable
        balance = newBalance;
    }
}
```

---

1. Local Variables

Declared **inside a method, constructor, or block.** Exist only while that block executes.

java

```
public class Calculator {

    public int add(int a, int b) {
```

```
        int result = a + b;       // local variable
        return result;
    }                             // result dies here

    public void process() {
        int x = 10;               // local to process()

        if (x > 5) {
            int y = 20;           // local to this if-block
            System.out.println(x + y);   // ✅ both visible here
        }

        System.out.println(x);    // ✅ x still visible
        System.out.println(y);    // ❌ compile error — y out of scope
    }
}
```

Key rules:

- **No default value** — must be initialized before use
- **No access modifiers** — can't be `public`, `private`, etc.
- **Lives on the stack** — created when block enters, destroyed when block exits

java

```
public void demo() {
    int x;                     // declared but NOT initialized
    System.out.println(x);    // ❌ compile error: variable x might not be initialize

    int y = 0;
    System.out.println(y);    // ✅ 0
}
```

---

2. Instance Variables

Declared **inside the class but outside any method.** Each object gets its **own independent copy.**

java

```
public class Student {

    // instance variables — each Student object has its own copy
    String name;
    int    age;
    double gpa;
```

```java
    public Student(String name, int age, double gpa) {
        this.name = name;
        this.age  = age;
        this.gpa  = gpa;
    }

    public void display() {
        System.out.println(name + " | Age: " + age + " | GPA: " + gpa);
    }
}
```

java

```java
Student s1 = new Student("Alice", 20, 3.9);
Student s2 = new Student("Bob",   22, 3.5);
Student s3 = new Student("Carol", 21, 3.7);

s1.display();    // Alice | Age: 20 | GPA: 3.9
s2.display();    // Bob   | Age: 22 | GPA: 3.5

// Each object owns its data — changing one doesn't affect others
s1.gpa = 4.0;
System.out.println(s2.gpa);    // 3.5 — unaffected
```

Memory:

```
HEAP
————
s1 → [ name="Alice", age=20, gpa=4.0 ]
s2 → [ name="Bob",   age=22, gpa=3.5 ]
s3 → [ name="Carol", age=21, gpa=3.7 ]
```

Default values when not initialized:

Type	Default
`int`, `short`, `byte`, `long`	`0`
`float`, `double`	`0.0`
`boolean`	`false`
`char`	`'\u0000'`
Object reference	`null`

java

```java
public class Demo {
    int     count;     // defaults to 0
    double  price;     // defaults to 0.0
    boolean active;    // defaults to false
    String  name;      // defaults to null

    void show() {
        System.out.println(count);    // 0
        System.out.println(price);    // 0.0
        System.out.println(active);   // false
        System.out.println(name);     // null
    }
}
```

---

3. Static Variables

Declared with `static` keyword. **One copy shared across ALL objects** of the class. Belongs to the class itself, not any instance.

java

```java
public class BankAccount {

    // static — ONE value shared by all BankAccount objects
    static int    totalAccounts = 0;
    static String bankName      = "JavaBank";

    // instance — each account has its own
    String owner;
    double balance;
    int    accountNumber;

    public BankAccount(String owner, double initialBalance) {
        totalAccounts++;                    // shared counter goes up
        this.accountNumber = totalAccounts; // assign current count as ID
        this.owner         = owner;
        this.balance       = initialBalance;
    }

    public static void showBankInfo() {     // static method
        System.out.println(bankName + " | Total Accounts: " + totalAccounts);
    }

    public void showAccount() {
        System.out.println("Acct #" + accountNumber + " | " + owner + " | ₹" + balan
```

```
    }
}
```

java

```
BankAccount a1 = new BankAccount("Alice", 50_000);
BankAccount a2 = new BankAccount("Bob",   30_000);
BankAccount a3 = new BankAccount("Carol", 75_000);

a1.showAccount();          // Acct #1 | Alice | ₹50000.0
a2.showAccount();          // Acct #2 | Bob   | ₹30000.0
a3.showAccount();          // Acct #3 | Carol | ₹75000.0

BankAccount.showBankInfo(); // JavaBank | Total Accounts: 3
```

Memory:

```
METHOD AREA (class-level)
──────────────────────────
BankAccount.totalAccounts = 3
BankAccount.bankName      = "JavaBank"

HEAP (per object)
──────────────────────────────────────────
a1 → [ owner="Alice", balance=50000, accountNumber=1 ]
a2 → [ owner="Bob",   balance=30000, accountNumber=2 ]
a3 → [ owner="Carol", balance=75000, accountNumber=3 ]
     (all three share the static variables above)
```

Side-by-side comparison

java

```
public class Employee {

    static String company   = "TechCorp";  // STATIC  — shared by all
    String        name;                    // INSTANCE — per object
    double        salary;                  // INSTANCE — per object

    public Employee(String name, double salary) {
        this.name   = name;
        this.salary = salary;
    }

    public void review() {
        double bonus = salary * 0.10;      // LOCAL — lives only here
```

```
        System.out.println(name + " bonus: ₹" + bonus);
    }
}
```

	Local	Instance	Static
Declared in	Method/block	Class body	Class body with `static`
Belongs to	Block execution	Each object	The class
Memory	Stack	Heap	Method Area
Lifetime	Block duration	Object lifetime	Program lifetime
Default value	❌ None (must init)	✅ Yes (0/null/false)	✅ Yes (0/null/false)
Access modifier	❌ Not allowed	✅ Allowed	✅ Allowed
Access via	Variable name	`object.field`	`ClassName.field`

Static variable — shared state in action

java

```java
public class Counter {
    static int count = 0;   // shared

    public Counter() {
        count++;
    }

    public static void main(String[] args) {
        System.out.println(Counter.count);   // 0

        new Counter();
        new Counter();
        new Counter();

        System.out.println(Counter.count);   // 3 — all three bumped same variable
    }
}
```

Shadowing — local hides instance

java

```java
public class Shadow {
    int x = 100;            // instance variable

    public void show(int x) {   // parameter shadows instance variable
        System.out.println(x);        // prints parameter value
        System.out.println(this.x);   // this.x explicitly accesses instance variabl
    }

    public static void main(String[] args) {
        new Shadow().show(999);
        // 999   ← local parameter
        // 100   ← instance variable via this.x
    }
}
```

Interview Questions

Q1. What is the difference between instance and static variables?

Instance variables have a **separate copy per object** — each object owns its own value. Static variables have **one copy for the entire class** — all objects read and write the same memory location. Instance variables live on the heap inside each object; static variables live in the Method Area.

```java
class Car {
    static int wheels = 4;   // all cars have 4 wheels — shared fact
    String     color;        // each car has its own color — per object
}
```

Q2. What is the default value of a local variable?

None. Unlike instance and static variables, local variables are **never given a default value** by the JVM. The compiler enforces that you initialize them before use — reading an uninitialized local is a **compile-time error**.

```java
void demo() {
    int x;
    System.out.println(x);    // ❌ compile error
}
```

Q3. Can you access a non-static variable inside a static method?

No — directly. A static method runs without any object, so there is no `this` and no instance to read fields from. You must pass an object reference:

```java
class Demo {
    int value = 10;

    static void show() {
        System.out.println(value);          // ❌ compile error

        Demo d = new Demo();
        System.out.println(d.value);        // ✅ via an object reference
    }
}
```

Q4. Can you access a static variable inside an instance method?

Yes — always. Instance methods have access to both instance and static members:

```java
class Demo {
    static int count = 5;
    int id = 1;

    void show() {
        System.out.println(count);   // ✅ static accessible in instance method
        System.out.println(id);      // ✅ instance variable
    }
}
```

Q5. Where are local, instance, and static variables stored in memory?

```
Local    → Stack  (inside the method's stack frame)
Instance → Heap   (inside the object)
Static   → Method Area  (class-level memory, loaded once per class)
```

Q6. What happens to static variables when objects are created and destroyed?

Q7. What is variable shadowing? How do you fix it?

Shadowing occurs when a local variable (or parameter) has the **same name as an instance variable**, hiding it inside that scope. Fix it using `this.variableName` to explicitly refer to the instance variable:

```java
class Box {
    int width = 10;

    void resize(int width) {        // parameter shadows instance field
        System.out.println(width);       // parameter: whatever was passed
        System.out.println(this.width);  // instance field: 10
        this.width = width;              // fix: assign param to instance field
    }
}
```

Q8. Can a static variable be declared inside a method?

No. `static` is a class-level concept. You cannot declare a static variable inside a method — it's a compile error:

```java
void demo() {
    static int x = 10;    // ❌ compile error — illegal modifier
}
```

Q9. What is the output?

```java
class Test {
    static int x = 5;

    public static void main(String[] args) {
        Test a = new Test();
        Test b = new Test();

        a.x = 10;
        System.out.println(b.x);    // ?
```

```
    }
}
```

Output: `10` — `x` is static, so `a.x` and `b.x` refer to the **same variable**. Changing it via `a` is visible via `b`. (Note: accessing static via instance reference like `a.x` is poor style — prefer `Test.x`.)

Q10. Can two local variables in the same method have the same name?

No — if they are in the same or overlapping scope. But a variable in an inner block can shadow one in an outer block in some cases:

java

```java
void demo() {
    int x = 10;
    int x = 20;       // ❌ compile error — duplicate in same scope

    int y = 10;
    {
        int y = 20;   // ❌ still compile error in Java — unlike C/C++
    }
}
```

Java is stricter than C/C++ — you **cannot** shadow a local variable with another local variable in a nested block.

Full summary

```
LOCAL
  ├── Inside method/block/constructor
  ├── Stack memory
  ├── No default — MUST initialize
  ├── No access modifiers
  └── Scope ends when block ends

INSTANCE
  ├── Inside class, outside methods
  ├── Heap memory (inside object)
  ├── Default values (0, null, false)
  ├── Access modifiers allowed
  └── One copy per object

STATIC
  ├── Inside class with static keyword
  ├── Method Area memory (class-level)
```

```
├── Default values (0, null, false)
├── Access modifiers allowed
├── One copy for ALL objects
└── Access via ClassName.variable
```

Constants (`final` keyword)

Constants & the `final` Keyword in Java

`final` is a **non-access modifier** that means **"this cannot be changed after assignment"**. It applies to variables, methods, and classes — with different meaning in each case.

`final` Variables — Constants

Once assigned, the value **cannot be reassigned:**

java

```java
public class Circle {

    final double PI = 3.14159265358979;   // constant

    public double area(double radius) {
        PI = 3.0;    // ❌ compile error: cannot assign a value to final variable PI
        return PI * radius * radius;    // ✅ reading is fine
    }
}
```

Three kinds of `final` variables

---

1. Final Local Variable

java

```java
public void process() {
    final int MAX_RETRY = 3;     // assigned once, never changed

    MAX_RETRY = 5;    // ❌ compile error

    for (int i = 0; i < MAX_RETRY; i++) {
```

```
        System.out.println("Attempt " + (i + 1));
    }
}
```

A final local variable doesn't need to be initialized at declaration — but must be initialized **before first use**, exactly once:

java

```
public void demo(boolean flag) {
    final int x;            // declared but not yet initialized

    if (flag) {
        x = 10;             // ✅ assigned in one branch
    } else {
        x = 20;             // ✅ assigned in other branch
    }

    System.out.println(x);  // ✅ safe — guaranteed to be assigned
    x = 99;                 // ❌ compile error — already assigned
}
```

---

2. Final Instance Variable

Must be initialized either at declaration or **inside the constructor** — never after:

java

```
public class Person {

    final String name;          // not initialized here
    final String nationalId;    // not initialized here
    final int    birthYear = 1995;  // initialized at declaration ✅

    public Person(String name, String nationalId) {
        this.name       = name;          // ✅ assigned in constructor
        this.nationalId = nationalId;    // ✅ assigned in constructor
    }

    public void birthday() {
        name = "New Name";    // ❌ compile error — already set
    }
}
```

java

```
Person p = new Person("Alice", "ID-001");
```

```
p.name = "Bob";    // ❌ compile error — final
```

---

3. Final Static Variable (true constants)

The most common pattern — `static final` together:

java

```java
public class MathConstants {

    // Convention: UPPER_SNAKE_CASE for constants
    public static final double PI          = 3.14159265358979;
    public static final double E           = 2.71828182845904;
    public static final int    MAX_INT     = Integer.MAX_VALUE;
    public static final String APP_VERSION = "2.1.0";
}
```

java

```java
// Access via class name — no object needed
double area = MathConstants.PI * radius * radius;
System.out.println(MathConstants.APP_VERSION);    // 2.1.0
```

Static final must be initialized at declaration or in a static block:

java

```java
public class Config {

    public static final String DB_URL;
    public static final int    TIMEOUT;

    // Static initializer block — runs once when class loads
    static {
        DB_URL  = "jdbc:mysql://localhost:3306/mydb";
        TIMEOUT = 30;
    }
}
```

`final` with Reference types — the important trap

`final` on a reference means the **reference cannot point to a different object** — but the object's contents can still change:

java

```java
```

```java
public class Demo {
    public static void main(String[] args) {

        final int[] numbers = {1, 2, 3};

        numbers[0] = 99;          // ✅ modifying contents — allowed
        System.out.println(numbers[0]);    // 99

        numbers = new int[]{4, 5, 6};     // ❌ compile error — reassigning referenc

        // ──────────────────────────────────────

        final StringBuilder sb = new StringBuilder("Hello");

        sb.append(" World");      // ✅ modifying object — allowed
        System.out.println(sb);   // Hello World

        sb = new StringBuilder(); // ❌ compile error — reassigning reference
    }
}
```

Mental model:

```
final int[] arr = {1, 2, 3}

STACK           HEAP
─────           ────
arr ──────────► [1, 2, 3]    ← contents CAN change
 ↑
 └── this arrow is final — cannot point elsewhere
```

Real-world constant patterns

App-wide constants class

java

```java
public class AppConstants {

    // prevent instantiation
    private AppConstants() { }

    // Server config
    public static final String HOST        = "api.myapp.com";
    public static final int    PORT        = 8080;
    public static final int    TIMEOUT_MS  = 5000;
```

```
    // Business rules
    public static final int    MAX_LOGIN_ATTEMPTS = 5;
    public static final double GST_RATE           = 0.18;
    public static final int    SESSION_EXPIRY_MIN = 30;

    // Error messages
    public static final String ERR_NOT_FOUND   = "Resource not found";
    public static final String ERR_UNAUTHORIZED = "Access denied";
}
```

java

```
// Use anywhere in the project
if (loginAttempts > AppConstants.MAX_LOGIN_ATTEMPTS) {
    System.out.println(AppConstants.ERR_UNAUTHORIZED);
}

double tax = amount * AppConstants.GST_RATE;
```

`final` Methods — cannot be overridden

java

```
public class Payment {

    // subclasses cannot override this — security-critical logic
    public final void processPayment(double amount) {
        validate(amount);
        deductBalance(amount);
        logTransaction(amount);
    }

    // these CAN be overridden
    protected void validate(double amount) {
        if (amount <= 0) throw new IllegalArgumentException("Invalid amount");
    }
}
```

java

```
public class UPIPayment extends Payment {

    @Override
    public void processPayment(double amount) {    // ❌ compile error
        // cannot override final method
    }
```

```
    @Override
    protected void validate(double amount) {       // ✅ not final, can override
        // UPI-specific validation
    }
}
```

When to use: Security-critical methods, template method patterns where the overall flow must stay fixed.

`final` Classes — cannot be extended

java

```
public final class ImmutablePoint {
    private final int x;
    private final int y;

    public ImmutablePoint(int x, int y) {
        this.x = x;
        this.y = y;
    }

    public int getX() { return x; }
    public int getY() { return y; }
}
```

java

```
public class MutablePoint extends ImmutablePoint {    // ❌ compile error
    // cannot extend a final class
}
```

Famous final classes in Java:

java

```
public final class String   { }     // java.lang.String
public final class Integer  { }     // java.lang.Integer
public final class Math     { }     // java.lang.Math
```

`String` is `final` — no one can subclass it and break its immutability contract.

`final` + `static` + `private` — full encapsulation

java

```java
public class DatabaseConfig {

    private static final String URL      = "jdbc:mysql://localhost/db";
    private static final String USER     = "admin";
    private static final String PASSWORD = "secret";
    private static final int    POOL     = 10;

    private DatabaseConfig() { }   // no instances

    public static String getUrl()  { return URL; }
    public static int    getPool() { return POOL; }
    // PASSWORD intentionally not exposed
}
```

`final` in summary — three contexts

java

```java
// 1. final VARIABLE — value cannot be reassigned
final int x = 10;
x = 20;    // ❌

// 2. final METHOD — cannot be overridden in subclass
public final void pay() { }

// 3. final CLASS — cannot be subclassed
public final class MyClass { }
```

Blank final variable

A `final` variable declared **without initialization** — must be assigned exactly once before use:

java

```java
public class Order {
    final String orderId;      // blank final — not yet set

    public Order(String id) {
        orderId = id;          // ✅ assigned in constructor
    }

    public Order() {
        // orderId not assigned   ❌ compile error
        // — blank final must be assigned in every constructor path
```

```
    }
}
```

Interview questions

Q1. Can a `final` variable be left uninitialized?

For **local** variables — yes, if initialized before first use. For **instance** final variables — must be initialized at declaration or in every constructor. For **static final** — at declaration or in a static block.

Q2. Does `final` make an object immutable?

No. `final` on a reference only prevents the reference from pointing to a different object. The object's internal state can still be mutated unless the class is designed to be immutable (all fields private + final, no setters):

java

```java
final List<String> list = new ArrayList<>();
list.add("hello");    // ✅ object mutated — final doesn't stop this
list = new ArrayList<>();  // ❌ reference reassignment — stopped by final
```

Q3. Can we initialize a `final` instance variable in a method?

No — only at declaration or in a constructor. Assigning in a regular method is a compile error:

java

```java
class Demo {
    final int x;

    void init() {
        x = 10;    // ❌ compile error — too late, not a constructor
    }
}
```

Q4. What is the difference between `final`, `finally`, and `finalize`?

```
final      → keyword: prevents reassignment / override / inheritance
finally    → try-catch block: code that always runs after try/catch
```

```
finalize() → deprecated method: called by GC before object is collected
```

```java
// final — modifier
final int x = 10;

// finally — exception handling
try {
    riskyOperation();
} catch (Exception e) {
    handle(e);
} finally {
    cleanup();    // always runs
}

// finalize — Object method (deprecated since Java 9)
@Override
protected void finalize() {
    // cleanup before GC — unreliable, avoid using
}
```

Q5. Why is `String` a `final` class?

Three reasons: **security** (no subclass can override `String` behavior and sneak malicious data into APIs expecting strings), **immutability** (the whole design of `String` relies on its value never changing — caching, hash codes, thread safety), and **performance** (the JVM heavily optimizes `String` because it knows exactly what it is).

Q6. Can a `final` method be overloaded?

Yes — overloading and overriding are different things. `final` blocks **overriding** (same signature in subclass), but you can freely **overload** (different parameters) even in the same class:

```java
public final void pay(double amount) { }      // final
public void pay(double amount, String upi) { } // ✅ overloaded — different params
```

Quick reference

Applied to	Effect
`final` local variable	Cannot be reassigned after first assignment
`final` instance variable	Must be set in constructor, never changed after
`final` static variable	Class-level constant, initialized once
`final` method	Cannot be overridden in subclasses
`final` class	Cannot be subclassed at all
`final` reference	Reference is fixed; object contents can still change

> **Rule of thumb:** use `final` by default for anything that shouldn't change. It documents intent, enables compiler optimizations, and prevents accidental mutation bugs.

Operators — arithmetic, logical, bitwise, ternary, shift
Most common Interview Questions asked

Operators in Java

An operator takes one or more operands and produces a result. Java has 6 major categories.

---

1. Arithmetic Operators

Basic math operations:

```java
int a = 17, b = 5;

System.out.println(a + b);   // 22  — addition
System.out.println(a - b);   // 12  — subtraction
System.out.println(a * b);   // 85  — multiplication
System.out.println(a / b);   // 3   — integer division (truncates!)
System.out.println(a % b);   // 2   — modulus (remainder)
```

Integer division truncates — does not round:

```java
```

```
System.out.println(7 / 2);       // 3   not 3.5
System.out.println(7.0 / 2);     // 3.5 — one double forces double math
System.out.println((double)7/2); // 3.5 — cast first, then divide
```

Modulus — not just for integers:

java

```
System.out.println(17 % 5);      // 2
System.out.println(-17 % 5);     // -2  — sign follows LEFT operand
System.out.println(17 % -5);     // 2
System.out.println(5.5 % 2.1);   // 1.3 — works on doubles too
```

Increment & Decrement:

java

```
int x = 5;

// Prefix — increment THEN use
System.out.println(++x);    // 6  (x becomes 6, then printed)
System.out.println(x);      // 6

// Postfix — use THEN increment
int y = 5;
System.out.println(y++);    // 5  (printed first, then y becomes 6)
System.out.println(y);      // 6
```

The classic trap:

java

```
int a = 5;
int b = a++ + ++a;
//      5   +  7    =  12
//      ↑ post: use 5, a→6
//             ↑ pre: a→7, use 7
System.out.println(b);    // 12
System.out.println(a);    // 7
```

---

2. Relational (Comparison) Operators

Always return `boolean`:

java

```
int a = 10, b = 20;
```

```
System.out.println(a == b);   // false — equal to
System.out.println(a != b);   // true  — not equal
System.out.println(a >  b);   // false — greater than
System.out.println(a <  b);   // true  — less than
System.out.println(a >= b);   // false — greater than or equal
System.out.println(a <= b);   // true  — less than or equal
```

`==` on objects — reference, not value:

java

```
String s1 = new String("hello");
String s2 = new String("hello");

System.out.println(s1 == s2);        // false — different objects on heap
System.out.println(s1.equals(s2));   // true  — same content

// String pool literals — same reference
String s3 = "hello";
String s4 = "hello";
System.out.println(s3 == s4);        // true  — same pool object
```

---

3. Logical Operators

Combine boolean expressions:

java

```
boolean a = true, b = false;

System.out.println(a && b);    // false — AND: both must be true
System.out.println(a || b);    // true  — OR:  at least one true
System.out.println(!a);        // false — NOT: inverts
```

Short-circuit evaluation — critical behavior:

java

```
// && stops at first false
int x = 0;
if (x != 0 && (10 / x > 1)) {     // second part NEVER evaluated
    System.out.println("yes");
}
// no ArithmeticException — && short-circuits at x != 0 (false)

// || stops at first true
String s = null;
if (s == null || s.isEmpty()) {    // second part NEVER evaluated if s==null
```

```
    System.out.println("empty");
}
// no NullPointerException — || short-circuits at s == null (true)
```

Non-short-circuit operators `&` and `|`:

java

```
int x = 0;
if (x != 0 & (10 / x > 1)) {   // ❌ both sides ALWAYS evaluated
    // throws ArithmeticException — division by zero
}
```

```
&&  →  short-circuit AND  — stops early, use this almost always
||  →  short-circuit OR   — stops early, use this almost always
&   →  non-short-circuit AND  — both sides always evaluated
|   →  non-short-circuit OR   — both sides always evaluated
```

---

4. Bitwise Operators

Operate on **individual bits** of integers:

java

```
int a = 12;   // binary: 1100
int b = 10;   // binary: 1010

//         1100
//         1010
//         ————
// AND  →  1000  = 8   (bit set only if BOTH bits are 1)
// OR   →  1110  = 14  (bit set if EITHER bit is 1)
// XOR  →  0110  = 6   (bit set if bits are DIFFERENT)
// NOT  → ~1100  = -13 (flips all bits, including sign bit)

System.out.println(a & b);    //  8
System.out.println(a | b);    // 14
System.out.println(a ^ b);    //  6
System.out.println(~a);       // -13
```

Visualized bit by bit:

```
a  =  0000 1100   (12)
b  =  0000 1010   (10)

a & b = 0000 1000  =  8   (AND  — 1 only where both are 1)
a | b = 0000 1110  = 14   (OR   — 1 where either is 1)
```

```
a ^ b = 0000 0110  =  6   (XOR  — 1 where they differ)
~a    = 1111 0011  = -13  (NOT  — all bits flipped)
```

Practical use — check/set/clear/toggle bits:

java

```java
int flags = 0b0000;

// Set bit 2 (turn ON)
flags = flags | (1 << 2);       // flags = 0100 = 4

// Check bit 2 (is it ON?)
boolean isSet = (flags & (1 << 2)) != 0;   // true

// Clear bit 2 (turn OFF)
flags = flags & ~(1 << 2);      // flags = 0000 = 0

// Toggle bit 2 (flip)
flags = flags ^ (1 << 2);       // flags = 0100 = 4 again
```

XOR swap — no temp variable:

java

```java
int x = 5, y = 9;
x = x ^ y;    // x = 5^9 = 12
y = x ^ y;    // y = 12^9 = 5  (original x)
x = x ^ y;    // x = 12^5 = 9  (original y)
System.out.println(x + " " + y);   // 9 5
```

---

5. Shift Operators

Shift bits left or right — equivalent to fast multiply/divide by powers of 2:

java

```java
int a = 8;    // binary: 0000 1000

// Left shift  << n  →  multiply by 2ⁿ
System.out.println(a << 1);    // 16  (8 × 2¹)
System.out.println(a << 2);    // 32  (8 × 2²)
System.out.println(a << 3);    // 64  (8 × 2³)

// Signed right shift >> n  →  divide by 2ⁿ (preserves sign)
System.out.println(a >> 1);    // 4   (8 / 2¹)
System.out.println(a >> 2);    // 2   (8 / 2²)
```

```
// Unsigned right shift >>> n  →  fills with 0 (ignores sign)
int neg = -8;
System.out.println(neg >> 1);   // -4  (sign preserved)
System.out.println(neg >>> 1);  // 2147483644  (fills with 0, huge positive)
```

Visualized:

```
a = 0000 1000  (8)

a << 1  →  0001 0000  = 16   (shift left,  fill right with 0)
a >> 1  →  0000 0100  =  4   (shift right, fill left with sign bit)

neg = 1111 1000  (-8)
neg >>  1  →  1111 1100  = -4           (sign bit (1) copied in)
neg >>> 1  →  0111 1100  = 2147483644   (0 filled in — not sign bit)
```

Why shift instead of multiply?

java

```
// These are equivalent but shift is marginally faster at hardware level
int x = 5;
System.out.println(x * 4);    // 20
System.out.println(x << 2);   // 20  — 5 × 2² = 20

System.out.println(x / 8);    // 0  (int division)
System.out.println(x >> 3);   // 0  — 5 / 2³ = 0

// Common in hashing, encoding, graphics, low-level protocols
int rgb   = (255 << 16) | (128 << 8) | 64;   // pack R,G,B into one int
int red   = (rgb >> 16) & 0xFF;               // extract red channel
int green = (rgb >> 8)  & 0xFF;               // extract green channel
int blue  =  rgb        & 0xFF;               // extract blue channel
```

---

6. Ternary Operator

Compact `if-else` in a single expression:

java

```
// Syntax: condition ? valueIfTrue : valueIfFalse

int age    = 20;
String result = (age >= 18) ? "Adult" : "Minor";
System.out.println(result);    // Adult

// Nested ternary — readable if kept simple
```

```java
int marks = 75;
String grade = (marks >= 90) ? "A"
             : (marks >= 75) ? "B"
             : (marks >= 60) ? "C"
             : "F";
System.out.println(grade);     // B
```

Ternary in expressions:

java

```java
int a = 10, b = 20;
int max = (a > b) ? a : b;                // 20
System.out.println("Max: " + max);

// Inline — avoid if too complex
System.out.println("Fee: ₹" + (isPremium ? 0 : 499));
```

Operator Precedence

Higher up = evaluated first:

```
Priority   Operators
────────   ──────────────────────────────
  1        ++  --  (postfix)
  2        ++  --  !  ~  (prefix/unary)
  3        *   /   %
  4        +   -
  5        <<  >>  >>>
  6        <   >   <=  >=  instanceof
  7        ==  !=
  8        &
  9        ^
  10       |
  11       &&
  12       ||
  13       ?:  (ternary)
  14       =   +=  -=  *=  /=  etc.
```

java

```java
int result = 2 + 3 * 4;          // 14 not 20 — * before +
int result = (2 + 3) * 4;        // 20 — parentheses override
boolean x  = true || false && false;  // true — && before ||
```

> **Rule:** When in doubt, use **parentheses**. Code clarity beats memorizing precedence.

Interview Questions

Q1. What is the output?

```java
int x = 10;
System.out.println(x++);   // ?
System.out.println(++x);   // ?
```

```
10  — postfix: print first (10), then increment (x=11)
12  — prefix:  increment first (x=12), then print
```

Q2. What is the output?

```java
int a = 5, b = 3;
System.out.println(a++ + ++b + a);
```

```
Step by step:
a++  → use 5, then a becomes 6
++b  → b becomes 4, use 4
a    → a is now 6
5 + 4 + 6 = 15
```

Q3. Difference between `&` and `&&`?

`&&` is short-circuit — if left side is `false`, right side is **never evaluated**. `&` always evaluates both sides. In practice, always use `&&` for boolean logic to avoid null pointer or divide-by-zero errors on the right side.

Q4. What is the output?

```java
System.out.println(10 / 3);      // ?
System.out.println(10 / 3.0);    // ?
System.out.println(10 % 3);      // ?
```

Q5. How do you check if a number is even using bitwise?

java

```java
// Bitwise AND with 1 — checks last bit
// Even numbers always have last bit = 0
// Odd  numbers always have last bit = 1

int n = 14;
if ((n & 1) == 0) {
    System.out.println("Even");   // Even
} else {
    System.out.println("Odd");
}
```

Faster than `n % 2 == 0` at the hardware level.

Q6. What is the output?

java

```java
int a = 5;
int b = a++ + a++;
System.out.println(a);   // ?
System.out.println(b);   // ?
```

```
a = 7   — incremented twice
b = 11  — first a++ uses 5 (a→6), second a++ uses 6 (a→7): 5+6=11
```

Q7. How to swap two numbers without a temp variable?

java

```java
// Using XOR
int x = 15, y = 27;
x = x ^ y;
y = x ^ y;
x = x ^ y;
System.out.println(x + " " + y);   // 27 15
```

```
// Using arithmetic
x = x + y;   // x = 42
y = x - y;   // y = 15
x = x - y;   // x = 27
// Risk: overflow for large numbers — XOR is safer
```

Q8. What is the output?

java

```
System.out.println(5 >> 1);    // ?
System.out.println(5 << 1);    // ?
System.out.println(-1 >>> 1);  // ?
```

```
2          — 5 / 2¹ = 2 (int division)
10         — 5 × 2¹ = 10
2147483647 — -1 is all 1s in binary; >>> fills with 0 → 0111...1111 =
Integer.MAX_VALUE
```

Q9. Can the ternary operator replace all `if-else`?

Technically yes for simple cases, but **no in practice**. Ternary is an **expression** (produces a value), while `if-else` is a **statement** (executes code). Ternary can't run multi-line logic, declare variables, or call void methods as a clean expression.

java

```
// ✅ Good use of ternary
String label = (score > 50) ? "Pass" : "Fail";

// ❌ Abuse — hard to read
String result = (a > b) ? (a > c) ? (a > d) ? "a" : "d" : "c" : "b";

// ✅ Prefer if-else for complex logic
if (a > b) {
    doThis();
    doThat();
} else {
    doOther();
}
```

Q10. What does `^` do — is it "power" in Java?

No — `^` is **XOR** (exclusive OR) in Java, not exponentiation. This is a common mistake:

java

```java
System.out.println(2 ^ 3);    // 1  — XOR of 2 and 3, NOT 2³=8!

// For power, use Math.pow()
System.out.println(Math.pow(2, 3));    // 8.0  ✅
```

Q11. What is the output?

java

```java
int x = 0;
boolean result = (x != 0) && (10 / x > 0);
System.out.println(result);   // ?
```

```
false — no exception!
&& short-circuits: (x != 0) is false, so right side never evaluated.
If & were used instead, ArithmeticException: / by zero.
```

Summary

```
Arithmetic   →  + - * / %          (watch int division & overflow)
Relational   →  == != > < >= <=    (use .equals() for objects)
Logical      →  && || !            (&& and || are short-circuit)
Bitwise      →  & | ^ ~            (operate on individual bits)
Shift        →  << >> >>>          (<< = ×2ⁿ, >> = ÷2ⁿ)
Ternary      →  ? :                (expression, not statement)

Key rules:
  ├── int/int = int  (always truncates)
  ├── postfix a++ returns old value, prefix ++a returns new
  ├── && short-circuits, & does not
  ├── ^ is XOR not power (use Math.pow for power)
  ├── >>> fills with 0, >> preserves sign bit
  └── final on reference ≠ immutable object
```